

Researching the heart of matter, Ernest Rutherford transformed our understanding of the atom, identifying its components and revealing its inner structure. He showed that atoms can disintegrate into smaller constituents and described two types of radiation generated by the process: gamma and beta radiation.

MILESTONES

RADIATION PARTICLES

Publishes his discovery of alpha and beta radiation and the disintegration of the elements in 1899.

ULTIMATE AWARD

Wins the Nobel Prize in Chemistry in 1908—and expresses surprise at being heralded as a chemist.

ATOMIC STRUCTURE

Proposes the “Rutherford model” of the atom in 1911, stating that electrons orbit a tiny nucleus.

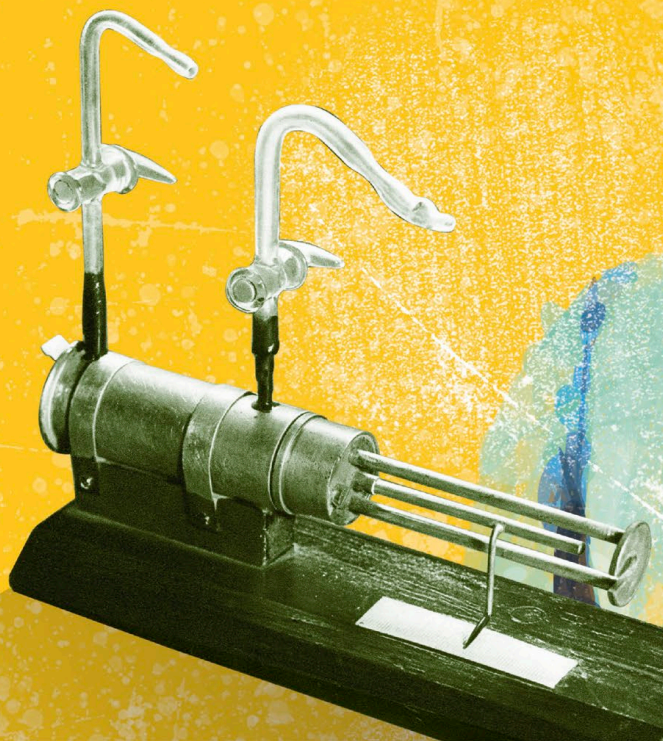
SPLITTING THE ATOM

Discovers the proton in 1919, and transforms one element into another: the process of nuclear fission.

Ernest Rutherford was born into a farming family in the village of Brightwater on the South Island of New Zealand. After winning a scholarship to Canterbury College (now the University of Canterbury), he gained three science degrees, then won an overseas scholarship and went to Cambridge University to work with the physicist J. J. Thomson. Impressed with the quality of Rutherford’s research, in 1898, Thomson recommended him for a professorship in physics at McGill University in Montreal, Canada. Aged only 27, Rutherford was accepted at McGill, and set sail for Canada.

It was in Montreal that Rutherford conducted the work that won him the Nobel Prize—for his “investigations into the disintegration of the elements.” Radioactivity had just been discovered by Henri Becquerel (see p.182), and soon afterward Thomson had identified the electron: a negatively charged subatomic particle. Up until then, atoms had been thought to be indivisible, but now it seemed that they were made up of smaller parts and that radioactivity was released as they decayed. Rutherford studied the radiation emitted by uranium and discovered that it was of two types, which he called alpha and beta radiation. He also found that alpha radiation was composed of positively charged particles and that

The first transmutation of the elements was achieved by Rutherford using this equipment (right). Nitrogen atoms were turned into oxygen by collision with alpha particles inside the tube. A window showed the protons emitted.



“I have broken the machine and **touched the ghost of matter.**”

Ernest Rutherford, 1917